



Wathiq: Smart e-Archive System

Graduation Project Report

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Abstract

In the era of rapid digital transformation, organizations face increasing challenges in managing large volumes of documents efficiently and securely. Traditional document management systems rely on Optical Character Recognition (OCR), which lacks semantic understanding.

This project presents **Wathiq**, an intelligent document management system that integrates OCR with Large Language Models (LLaMA). The system extracts text and understands document content to generate metadata, summaries, and keywords.

It supports multi-institution environments with Role-Based Access Control (RBAC) and advanced search using Elasticsearch. Evaluation using CER and WER shows improved performance.

Contents

Abstract	i
1 General Introduction	2
1.1 Project Overview and Background	2
1.2 Problem Statement	2
1.3 Project Objectives	2
1.4 Scope, Constraints, and Assumptions	3
1.4.1 Scope	3
1.4.2 Constraints	3
1.4.3 Assumptions	3
1.5 Project Contribution	3
1.6 Research Methodology	3
1.7 Document Structure	3
1.8 Definitions	4

CHAPTER 1

GENERAL INTRODUCTION

Chapter 1

General Introduction

1.1 Project Overview and Background

The rapid advancement of digital transformation has significantly increased the need for intelligent document management systems. Traditional paper-based systems are inefficient and prone to errors, while OCR-based systems lack semantic understanding.

The **Wathiq system** integrates OCR and LLaMA to provide both text extraction and semantic understanding. The system automatically generates metadata, summaries, and keywords, improving document processing efficiency.

It supports multi-institution environments with secure data isolation and Role-Based Access Control (RBAC). Elasticsearch is used to enable advanced search capabilities.

1.2 Problem Statement

Existing document management systems rely on OCR only, which limits their ability to understand document content. This leads to poor classification, weak metadata extraction, and inefficient search.

1.3 Project Objectives

Main Objective

- Develop an intelligent document management system integrating OCR and LLaMA.

Sub Objectives

- Extract text using OCR
- Analyze documents using LLaMA

- Generate metadata and summaries
- Support multi-institution systems
- Implement RBAC
- Enable advanced search

1.4 Scope, Constraints, and Assumptions

1.4.1 Scope

A web-based system for document management and processing.

1.4.2 Constraints

- Limited computational resources
- Time constraints

1.4.3 Assumptions

- Users have basic technical knowledge

1.5 Project Contribution

This project introduces a hybrid approach combining OCR and LLaMA for intelligent document understanding and retrieval.

1.6 Research Methodology

The system is evaluated using real datasets and performance metrics such as CER and WER.

1.7 Document Structure

- Chapter 1: Introduction
- Chapter 2: Project Management
- Chapter 3: Literature Review
- Chapter 4: System Analysis

- Chapter 5: System Design
- Chapter 6: Implementation
- Chapter 7: Testing
- Chapter 8: Results
- Chapter 9: Conclusion

1.8 Definitions

- OCR: Optical Character Recognition
- LLaMA: Large Language Model
- RBAC: Role-Based Access Control
- Metadata: Structured document information